

AD7190

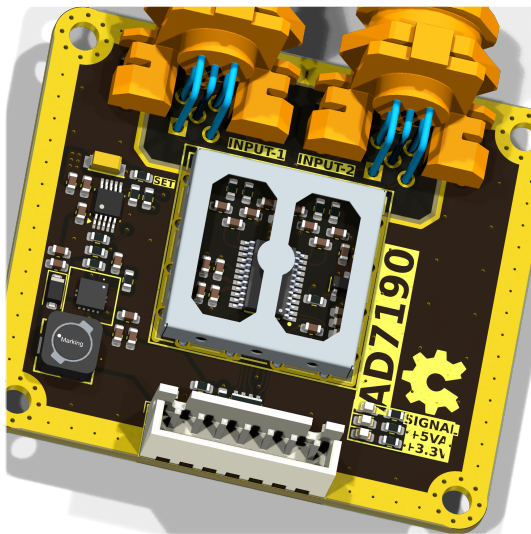
Variant:

2026-09-29

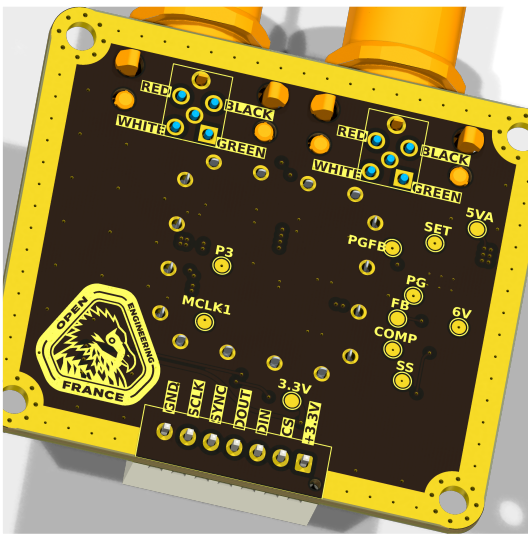
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

OPEN\_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.  
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 29-Sep-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

+3.3V host -> TPS61086  
6 V -> LT3042 +5VA (AVDD + excitation)

DEBUG NOTE:

TP on every rail.  
Wait >= 600 ms (PG) before ADC init.

DESIGN NOTE:

SPI mode 3,  
SCLK <= 5 MHz,  
RDY on DOUT.

DESIGN NOTE:

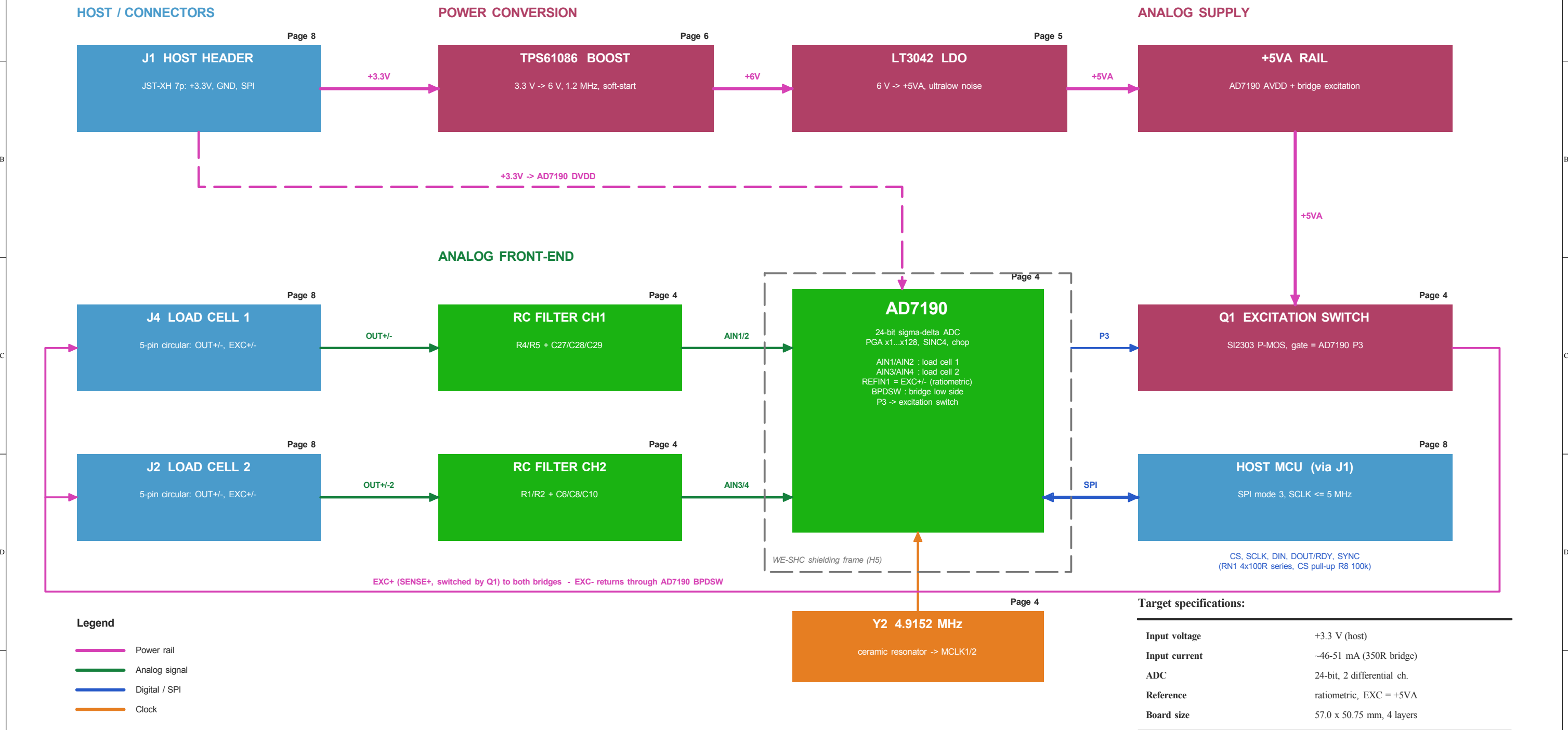
REFIN1 = bridge excitation:  
ratiometric.

LAYOUT NOTE:

AFE under the WE-SHC shield H5,  
boost SW node away.

|                  |                                     |                              |                     |                              |                      |
|------------------|-------------------------------------|------------------------------|---------------------|------------------------------|----------------------|
|                  | Comments:                           | Company:<br>OPEN THRUST LAB  |                     | Variant:                     | Git Hash:<br>b3867f9 |
|                  | Title:<br>AD7190                    | Board Name:<br>AD7190        |                     | Project Name:<br>OPEN_WEIGHT |                      |
|                  | File Name:<br>OPEN_WEIGHT.kicad_sch | Designer:<br>SIEBERT DIMITRY | Date:<br>2026-09-29 | Revision:<br>+ (Unreleased)  |                      |
| Sheet Path:<br>/ |                                     | Reviewer:                    | Size:<br>A3         | Sheet:<br>1 of 10            |                      |

# [2] Block Diagram

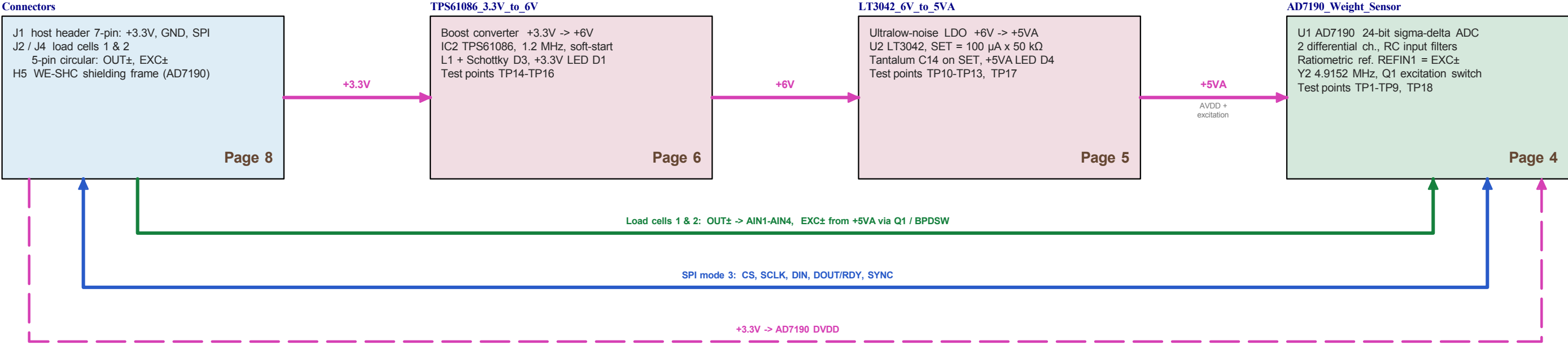


|                                                                                       |           |                         |                 |                 |          |               |           |                |  |
|---------------------------------------------------------------------------------------|-----------|-------------------------|-----------------|-----------------|----------|---------------|-----------|----------------|--|
|  | Comments: |                         | Company:        |                 | Variant: |               | Git Hash: |                |  |
|                                                                                       |           |                         | OPEN THRUST LAB |                 |          |               | b3867f9   |                |  |
|                                                                                       |           |                         | Board Name:     |                 |          | Project Name: |           |                |  |
|                                                                                       |           |                         | AD7190          |                 |          | OPEN_WEIGHT   |           |                |  |
| Sheet Title:                                                                          |           | File Name:              |                 | Designer:       |          | Date:         |           | Revision:      |  |
| Block Diagram                                                                         |           | Block Diagram.kicad_sch |                 | SIEBERT DIMITRY |          | 2026-09-29    |           | + (Unreleased) |  |
| Sheet Path:                                                                           |           |                         |                 | Reviewer:       |          | Size:         |           | Sheet:         |  |
| /Block Diagram/                                                                       |           |                         |                 |                 |          | A3            |           | 2 of 10        |  |

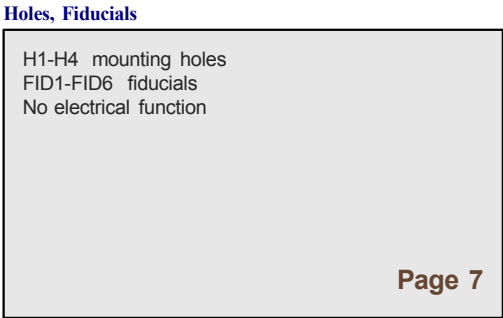
# [3] Project Architecture

Sheet hierarchy and signal flow. Click a block or its page number to open the sheet.

## HOST INTERFACE



## MECHANICAL



## LEGEND

- Power rail
- Secondary supply
- Analog signal
- Digital / SPI

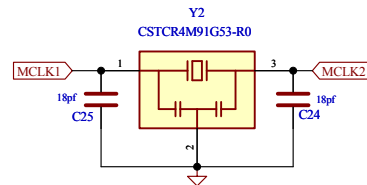
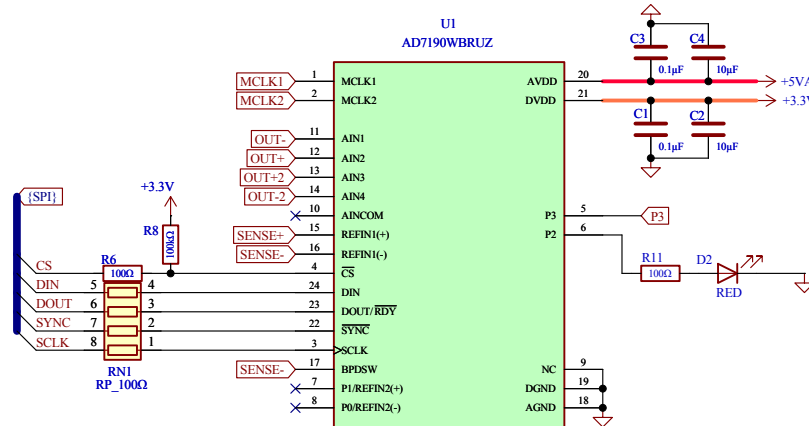
## DOCUMENT INDEX

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- [3] Project Architecture
- [4] AD7190 Weight Sensor
- [5] LT3042 LDO 6V -> 5VA
- [6] TPS61086 Boost 3V3 -> 6V
- [7] Holes, Fiducials
- [8] Connectors
- [9] Power Sequencing
- [10] Revision History



|                                       |                                              |                              |                              |                             |
|---------------------------------------|----------------------------------------------|------------------------------|------------------------------|-----------------------------|
| Comments:                             | Company:<br>OPEN THRUST LAB                  |                              | Variant:                     | Git Hash:<br>b3867f9        |
|                                       | Board Name:<br>AD7190                        |                              | Project Name:<br>OPEN_WEIGHT |                             |
| Sheet Title:<br>Project Architecture  | File Name:<br>Project Architecture.kicad_sch | Designer:<br>SIEBERT DIMITRY | Date:<br>2026-09-29          | Revision:<br>+ (Unreleased) |
| Sheet Path:<br>/Project Architecture/ |                                              | Reviewer:                    | Size:<br>A3                  | Sheet:<br>3 of 10           |

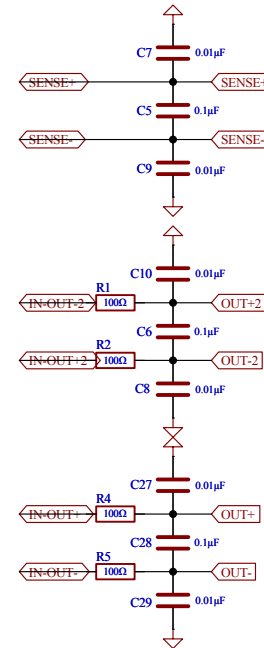
# [4] AD7190 - Weight Sensor



**ADC FOR  
weighing sensor**

## DESIGN NOTE:

Ref Condo for quartz  
Murata GRM1885C1H220JA01D  
(22 pF, 50 V, C0G, 0603, ±5%)



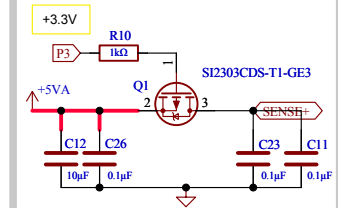
Calcul de filtre :  $f_c = 1 / (2\pi \times R \times C) = X \text{ Hz}$

## DESIGN NOTE:

Ref Condo for Filtre  
Murata GRM1885C1H103JA01D (10 nF, 50 V, C0G, 0603, ±5%)  
KEMET C0603R104K5RACTU (100 nF, 50 V, X7R, 0603, ±10%)

Ref Resistor for Filtre  
TNPW0603100RBEEA (100 Ω, 0.1 W, thin film, 0603, ±0.1%, 25 ppm/°C)

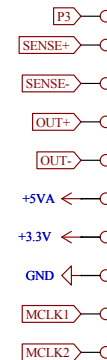
**FILTRE**



## DESIGN NOTE:

IC1 is a P-MOS high-side switch used to enable/disable the load cell excitation. must disabled during calibration

**DRIVE GATE**



**TEST POINT**

## About

The AD7190 is a low noise, complete analog front end for high precision measurement applications. It contains a low noise, 24-bit sigma-delta ( $\Sigma\Delta$ ) analog-to-digital converter (ADC). The on-chip low noise gain stage means that signals of small amplitude can be interfaced directly to the ADC. The device can be configured to have two differential inputs or four pseudo differential inputs. The on-chip channel sequencer allows several channels to be enabled, and the AD7190 sequentially converts on each enabled channel. This simplifies communication with the part. The on-chip 4.92 MHz clock can be used as the clock source to the ADC or, alternatively, an external clock or crystal can be used. The output data rate from the part can be varied from 4.7 Hz to 4.8 kHz. The device has two digital filter options. The choice of filter affects the rms noise/noise-free resolution at the programmed output data rate, the settling time, and the 50 Hz/60 Hz rejection.



## Comments:

AD7190  
CSTCR4M91G53-R0  
AD7190\_TEST\_BOARD CN

## Company:

OPEN THRUST LAB

## Board Name:

AD7190

## Sheet Title:

AD7190 - Weight Sensor

## Sheet Path:

/Project Architecture/AD7190\_Weight\_Sensor/

## File Name:

AD7190 - Weight Sensor.kicad\_sch

## Designer:

SIEBERT DIMITRY

## Reviewer:

## Variant:

## Git Hash:

b3867f9

## Project Name:

OPEN\_WEIGHT

## Date:

2026-09-29

## Revision:

+(Unreleased)

## Size:

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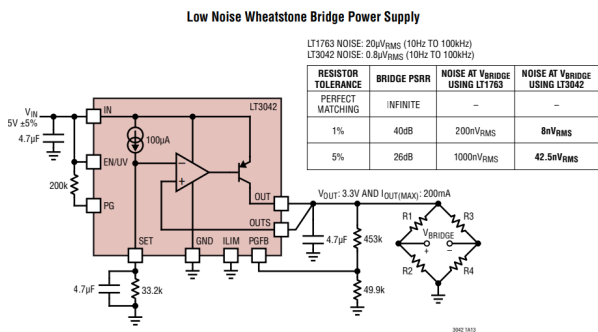
## Sheet:

4 of 10

# [5] LT3042 - LDO

## 6V to 5V Analog

### TYPICAL APPLICATIONS



IN (pins 1-2): LT3042 power input (6V), to be decoupled with CIN.

EN/UV (pin 3): enables/disables the regulator and allows an UVLO threshold (often tied to IN for "always ON").

PG (pin 4): open-collector Power-Good output (requires a pull-up resistor).

ILIM (pin 5): current limit programming/monitoring (often tied to GND if unused).

PGFB (pin 6): input that sets the Power-Good threshold via a resistor divider from VOUT.

SET (pin 7): sets the output voltage via RSET (and CSET reduces noise / provides soft-start).

GND (pin 8): circuit ground reference.

OUTS (pin 9): Kelvin output sense (to be connected to the + of COUT / load point).

OUT (pin 10): regulated output (5V), to be decoupled with COUT.

EPAD (exposed pad): ground + thermal dissipation (to be soldered to the GND plane).

#### DESIGN NOTE:

Ref Condo for SET PIN (important pour les vibrations P16)  
KYOCERA AVX TAJA225K016RNJ (2.2 µF, 16 VDC,  
condensateur tantale solide SMD moulé, 1206 / 3216 (A case),  
±10%, ESR 6.5 Ω).

$V_{OUT} \approx I_{SET} \times R_{SET}$   
 $I_{SET} \approx 100 \mu A$

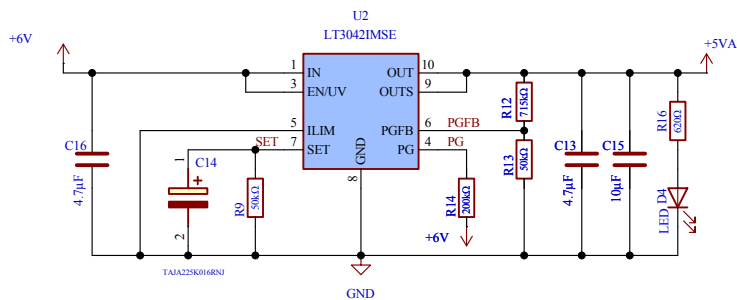
Donc :  
 $R_{SET} \approx V_{OUT} / I_{SET}$   
 $R_{SET} \approx 5.0 V / 100 \mu A \approx 50 k\Omega$   
(=> valeur standard recommandée : 49.9 kΩ)

$V_{OUT}(PG\_THRESHOLD) \approx 0.3 V \times (1 + R_{PG2}/R_{PG1})$   
 $PGFB\_THRESHOLD \approx 0.3 V$

Donc :  
 $R_{PG2} \approx R_{PG1} \times (V_{OUT}(PG)/0.3 V - 1)$

Exemple (PG OK à ~4.8 V) :  
Choix  $R_{PG1} \approx 49.9 k\Omega$   
 $R_{PG2} \approx 49.9 k\Omega \times (4.8/0.3 - 1) \approx 748.5 k\Omega$   
(=> valeur standard recommandée : 750 kΩ)

### Design guide line



### Convertisseur 5VA

#### About

The LT®3042 is a high performance low dropout linear regulator featuring LTC's ultralow noise and ultrahigh PSRR architecture for powering noise sensitive RF applications. Designed as a precision current reference followed by a high performance voltage buffer, the LT3042 can be easily paralleled to further reduce noise, increase output current and spread heat on the PCB. The device supplies 200mA at a typical 350mV dropout voltage. Operating quiescent current is nominally 2mA and drops to <<1µA in shutdown. The LT3042's wide output voltage range (0V to 15V) while maintaining unitygain operation provides virtually constant output noise, PSRR, bandwidth and load regulation, independent of the programmed output voltage. Additionally, the regulator features programmable current limit, fast start-up capability and programmable power good to indicate output voltage regulation. The LT3042 is stable with a minimum 4.7µF ceramic output capacitor. Built-in protection includes reverse battery protection, reverse current protection, internal current limit with foldback and thermal limit with hysteresis. The LT3042 is available in thermally enhanced 10-Lead MSOP and 3mm × 3mm DFN package

### TEST POINT



#### Comments:

LT3042  
Analog Devices

#### Sheet Title:

LT3042 - LDO 6V to 5V Analog

#### Sheet Path:

/Project Architecture/LT3042\_6V\_to\_5VA/

#### Company:

OPEN THRUST LAB

#### Board Name:

AD7190

#### File Name:

LT3042 - LDO 6V to 5VA.kicad\_sch

#### Designer:

SIEBERT DIMITRY

#### Reviewer:

#### Variant:

#### Project Name:

OPEN\_WEIGHT

#### Date:

2026-09-29

#### Size:

A3

#### Git Hash:

b3867f9

#### Revision:

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#### Revision:

(Unreleased)

#### Sheet:

5

of 10

# [6] TPS61086 - Boost 3.3V to 6V

COMP (pin 1): error amplifier compensation pin; connect the Rcomp + Ccomp network to AGND to stabilize the loop (do not route it near SW).

FB (pin 2): feedback input; connect it to the R1/R2 resistor divider from VOUT (after the diode) to set the output voltage (e.g. 6 V).

EN (pin 3): enable / shutdown; high level = active, low level = off. Often tied to IN for "always ON", or driven by a GPIO.

AGND (pin 4): analog ground (reference for sensitive signals: FB/COMP/SS/MODE). Keep it "clean" and connect it to the GND plane in a controlled way.

PGND (pin 5): power ground (return path for switching currents). Connect it to the GND plane with a short/wide copper area close to the power capacitors.

SW (pins 6-7): switch node; connect it to L and the Schottky diode. This is the noisiest node → keep the trace short, minimize the loop area, and keep it away from FB/COMP.

IN (pin 8): power input (2.3-6 V); decouple it as close as possible with Cby ~1 μF + Cin (e.g. 10 μF) to GND.

MODE (pin 9): mode selection; MODE = High → Forced PWM (more stable frequency, better for noise), MODE = Low → PFM (better efficiency at light load).

SS (pin 10): soft-start; add a capacitor (C<sub>SS</sub>) to limit inrush current at startup. Left open = no soft-start.

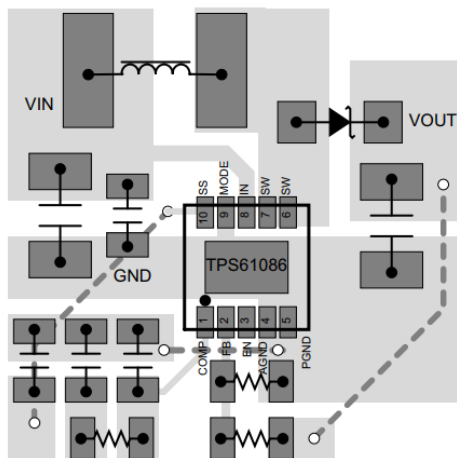
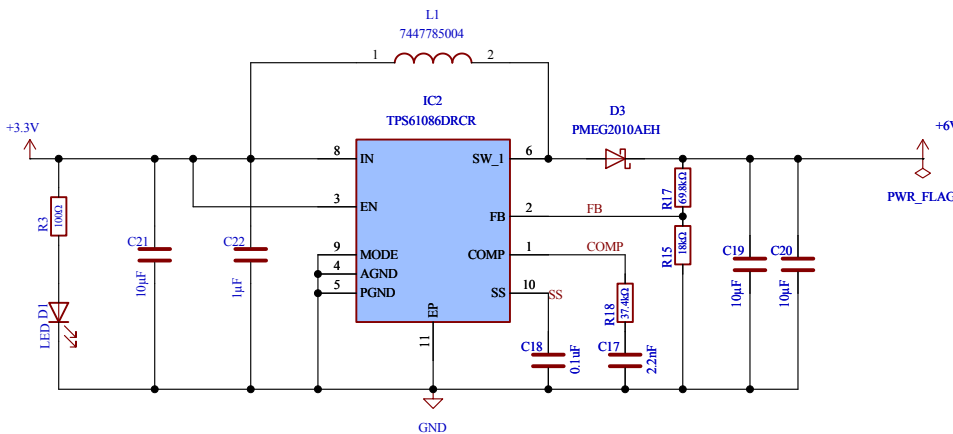


Figure 26. TPS61086 Layout Example

Pin usage and layout example



Boost 3.3 V -> 6.04 V

## About

The TPS61086 device is a high-frequency, high-efficiency DC-to-DC converter with an integrated 2.0-A, 0.13-Ω power switch capable of providing an output voltage up to 18.5 V. The implemented boost converter is based on a fixed frequency of 1.2-MHz, pulse-width-modulation (PWM) controller that allows the use of small external inductors and capacitors and provides fast transient response.

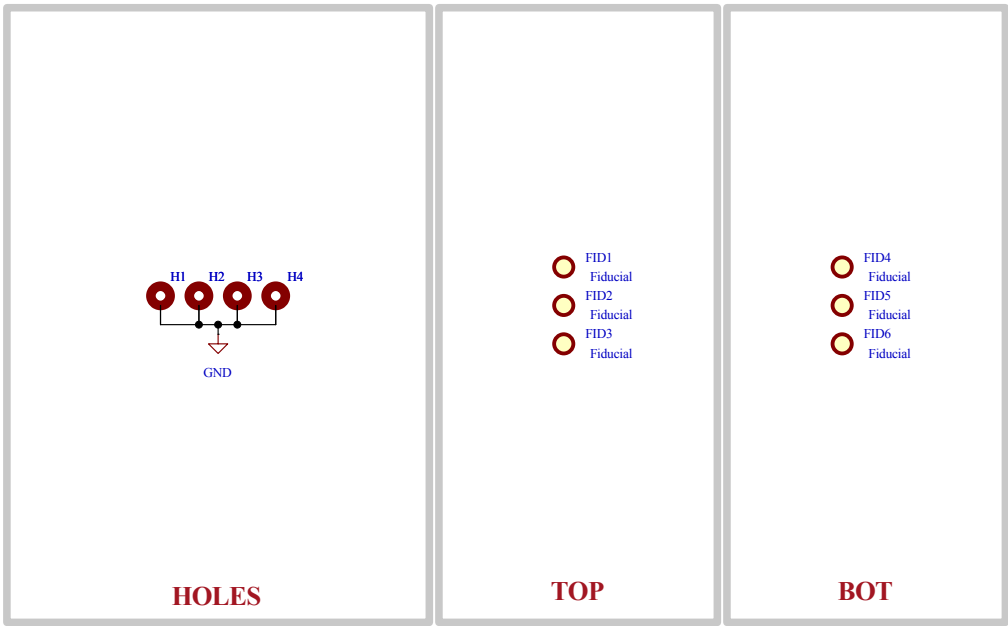
At light-load, the device can operate in Power Save Mode with pulse-frequency-modulation (PFM) to improve the efficiency while keeping a low-output voltage ripple. For very noise-sensitive applications, the device can be forced to PWM Mode operation over the entire load range by pulling the MODE pin high. The external compensation allows optimizing the application for specific conditions. A capacitor connected to the soft-start pin minimizes inrush current at start-up.

Test points



|                                                        |                                      |                 |               |                |           |
|--------------------------------------------------------|--------------------------------------|-----------------|---------------|----------------|-----------|
| Comments:                                              | Company:                             |                 | Variant:      |                | Git Hash: |
| References:                                            | OPEN THRUST LAB                      |                 |               |                | b3867f9   |
| TI TPS61086 datasheet<br>L1 7447785004, D3 PMEG2010AEH | Board Name:                          |                 | Project Name: |                |           |
|                                                        | AD7190                               |                 | OPEN_WEIGHT   |                |           |
| Sheet Title:                                           | File Name:                           | Designer:       | Date:         | Revision:      |           |
| TPS61086 - Boost 3.3V to 6V                            | TPS61086 - Boost 3V3 to 6V.kicad_sch | SIEBERT DIMITRY | 2026-09-29    | + (Unreleased) |           |
| Sheet Path:                                            | Reviewer:                            |                 | Size:         | Sheet:         |           |
| /Project Architecture/TPS61086_3.3V_to_6V/             |                                      |                 | A3            | 6 of 10        |           |

# [7] Holes, Fiducials



|  |                                                                                                                        |                                                  |                                     |                                     |                                    |
|--|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|
|  | <b>Comments:</b><br><b>References:</b><br>4 x M3 (3.2 mm) plated holes to GND<br>FID1-FID6 not fitted on this revision | <b>Company:</b><br>OPEN THRUST LAB               |                                     | <b>Variant:</b>                     | <b>Git Hash:</b><br>b3867f9        |
|  | <b>Sheet Title:</b><br>Holes, Fiducials                                                                                | <b>Board Name:</b><br>AD7190                     |                                     | <b>Project Name:</b><br>OPEN_WEIGHT |                                    |
|  | <b>Sheet Path:</b><br>/Project Architecture/Holes, Fiducials/                                                          | <b>File Name:</b><br>Holes - Fiducials.kicad_sch | <b>Designer:</b><br>SIEBERT DIMITRY | <b>Date:</b><br>2026-09-29          | <b>Revision:</b><br>+ (Unreleased) |
|  |                                                                                                                        |                                                  | <b>Reviewer:</b>                    | <b>Size:</b><br>A3                  | <b>Sheet:</b><br>7 of 10           |

[8] Connectors

DESIGN NOTE:  
J1 host header (JST-XH, 7 pins):  
1 +3.3V 2 CS 3 DIN 4 DOUT/RDY  
5 SYNC 6 SCLK 7 GND

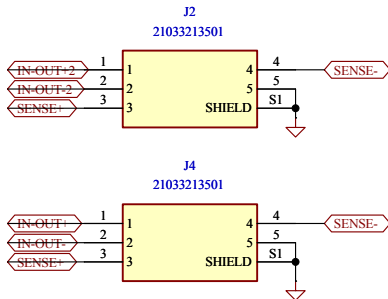
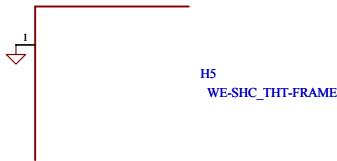
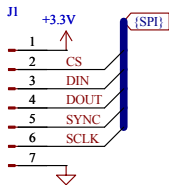
DESIGN NOTE:  
J2/J4 load cell (5 pins):  
1 OUT, 2 OUT (signal pair), 3 EXC+ (SENSE+),  
4 EXC- (SENSE-/BPDSW), 5 GND/shield.  
Both channels read V(pin2) - V(pin1):  
fix the sign in firmware if needed.

LAYOUT NOTE:  
H5 shielding frame covers the AD7190  
front-end; connector shells tied to GND.

Pinout and notes



Power flags: +3.3V and GND come from the host connector J1,  
(J2/J4 21033213501, H5 WE-SHC).



Host and load-cell connectors



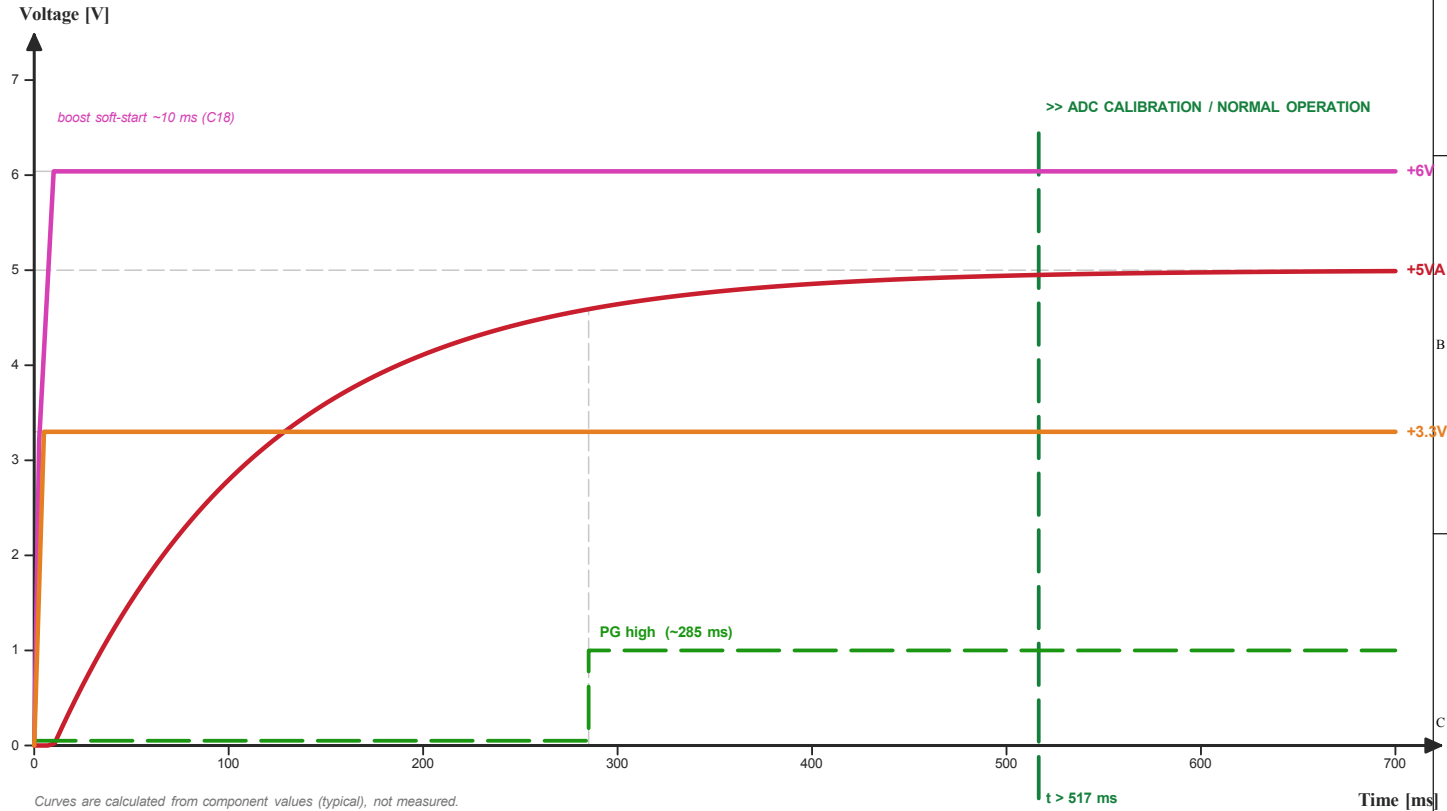
|                                                                                                 |                                           |                                     |                                     |                                    |
|-------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|
| <b>Comments:</b><br><b>References:</b><br>JST XH B7B-XH-AM (J1)<br>J2/J4 21033213501, H5 WE-SHC | <b>Company:</b><br>OPEN THRUST LAB        |                                     | <b>Variant:</b>                     | <b>Git Hash:</b><br>b3867f9        |
|                                                                                                 | <b>Board Name:</b><br>AD7190              |                                     | <b>Project Name:</b><br>OPEN_WEIGHT |                                    |
| <b>Sheet Title:</b><br>Connectors                                                               | <b>File Name:</b><br>Connectors.kicad_sch | <b>Designer:</b><br>SIEBERT DIMITRY | <b>Date:</b><br>2026-09-29          | <b>Revision:</b><br>+ (Unreleased) |
| <b>Sheet Path:</b><br>/Project Architecture/Connectors/                                         |                                           | <b>Reviewer:</b>                    | <b>Size:</b><br>A3                  | <b>Sheet:</b><br>8 of 10           |

# [9] Power - Sequencing

| NAME  | SOURCE       | LEVEL        |
|-------|--------------|--------------|
| +3.3V | Host (J1)    | 3.3 V (host) |
| +6V   | TPS61086 IC2 | 6.04 V       |
| +5VA  | LT3042 U2    | 5.00 V       |
| PG    | LT3042 U2    | > 4.59 V     |

Values from the schematic:  
+6V = 1.238 V x (1 + R17/R15) = 1.238 x (1 + 69.8k/18k)  
+5VA = ISET x RSET = 100 uA x 50k (R9)  
PG threshold = 0.3 V x (1 + R12/R13) = 4.59 V  
LT3042 soft-start: tau = RSET x CSET = 50k x 2.2 uF  
= 110 ms -> +5VA at 99 % after ~550 ms

DESIGN NOTE:  
The LT3042 SET capacitor (C14 = 2.2 uF) gives a 110 ms time constant. Firmware must wait >= 600 ms after power-up (or until PG, TP17) before the AD7190 reset and internal calibration. To start faster, reduce C14 (e.g. 0.47 uF -> tau = 23.5 ms) at the cost of slightly more noise.



|  |                                     |                                            |                              |                              |                             |
|--|-------------------------------------|--------------------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                           | Company:<br>OPEN THRUST LAB                |                              | Variant:                     | Git Hash:<br>b3867f9        |
|  |                                     | Board Name:<br>AD7190                      |                              | Project Name:<br>OPEN_WEIGHT |                             |
|  | Sheet Title:<br>Power - Sequencing  | File Name:<br>Power - Sequencing.kicad_sch | Designer:<br>SIEBERT DIMITRY | Date:<br>2026-09-29          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:<br>/Power - Sequencing/ |                                            | Reviewer:                    | Size:<br>A4                  | Sheet:<br>9 of 10           |

# [10] Revision History

## Version Unreleased

### Added

- AD7190 front-end, 2 load-cell inputs
- +3.3V -> TPS61086 6 V -> LT3042 +5VA
- Self-contained libs and 3D models (lib/)
- JLCPCB 4-layer rules and stackup
- LTspice simulations (Simulation/LTspice)

### Changed

- Renamed to OPEN\_WEIGHT, KiCad 10

### Fixed

- Connectors not linked to ADC in schematic
- AD7190 solder-mask bridges

|  |                                   |                                          |                              |                              |                             |
|--|-----------------------------------|------------------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                         | Company:<br>OPEN THRUST LAB              |                              | Variant:                     | Git Hash:<br>b3867f9        |
|  |                                   | Board Name:<br>AD7190                    |                              | Project Name:<br>OPEN_WEIGHT |                             |
|  | Sheet Title:<br>Revision History  | File Name:<br>Revision History.kicad_sch | Designer:<br>SIEBERT DIMITRY | Date:<br>2026-09-29          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:<br>/Revision History/ |                                          | Reviewer:                    | Size:<br>A4                  | Sheet:<br>10 of 10          |